

Intel ipu7 otocam driver User Manual

Revision	Description	Release Date
V0.02.00	Initial Release	2026/07/01
V0.03.00	Support otocam260isp/otocam271isp/otocam276isp	2026/07/31
V0.03.01	Support otocam274isp/otocam278isp	2026/08/14
V0.03.02	Support remove & insert driver	2026/08/19

1. Get started Guide:

- While setup the build environment on the new ipu7 machine, do items from here:
 - follow the steps in the doc "858119_PantherLakeH_Linux_GetStartedGuide_0.81.pdf" and fix the steps of session 2.3.2 for building the linux kernel:
 - follow the steps:
 - (a) `sudo git clone https://github.com/intel/linux-kernel-overlay.git`
 - (b) `cd linux-kernel-overlay`
 - (c) `sudo ./build.sh -r yes -t mainline-tracking-overlay-pre-prod-v6.17-ubuntu-251118T134731Z -b 1000 -c preempt-rt`

Once the build is successful, the following Debian packages will be generated under the `linux-kernel-overlay` directory:

- `linux-headers-<version>*_amd64.deb`
- `linux-image-<version>*_amd64.deb`

2. Driver installation :

- (a) retrieve the otocam-intel-driver:
 - `$cd /home/user/otocam-intel-driver-vx.xx.xx/script/`
 - `$chmod 777 *.sh`
 - `./install_image.sh` (update .deb from step1)
 - `./install_ko.sh`
 - `./install_json.sh`
- While update new ko or xml, do items from here:
 - (b) copy the ko & xml to the intel system:
 - `$cd /home/user/otocam-intel-driver-vx.xx.xx/script/`
 - `./install_ko.sh`
 - `./install_json.sh`

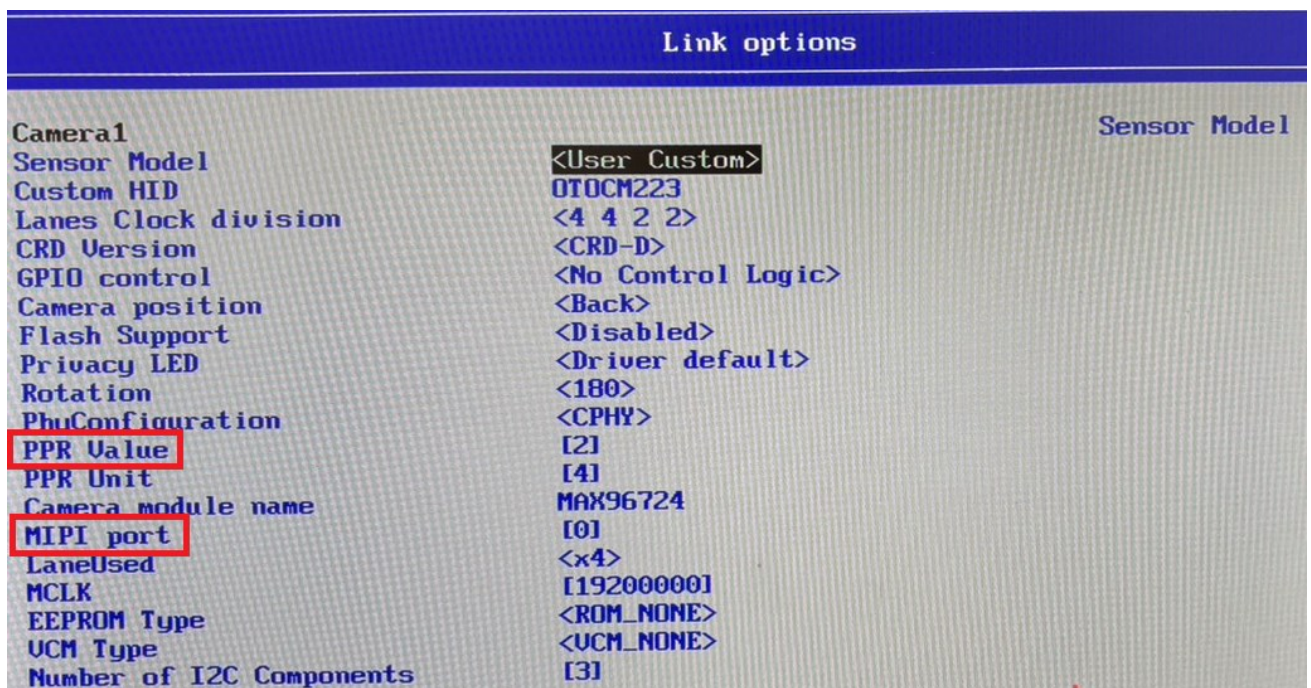
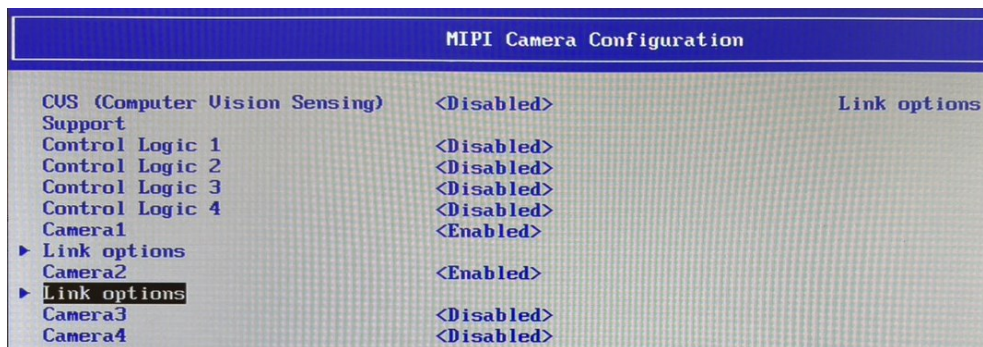
3.Steps to execute sensor preview:

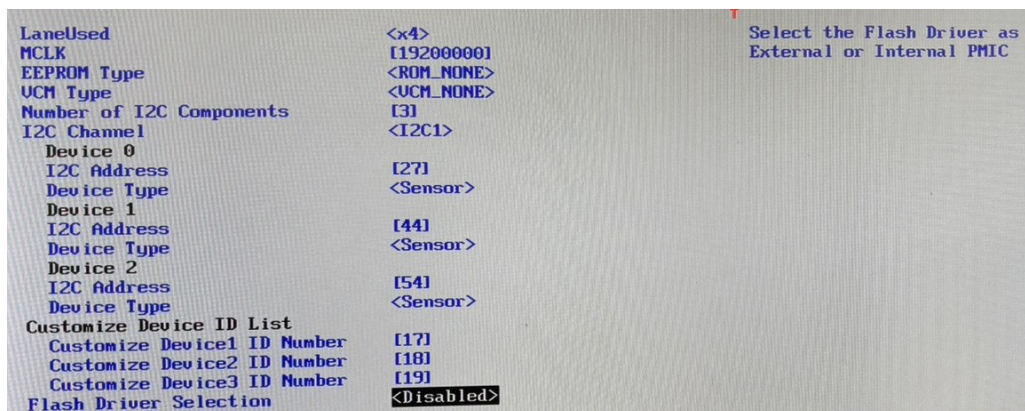
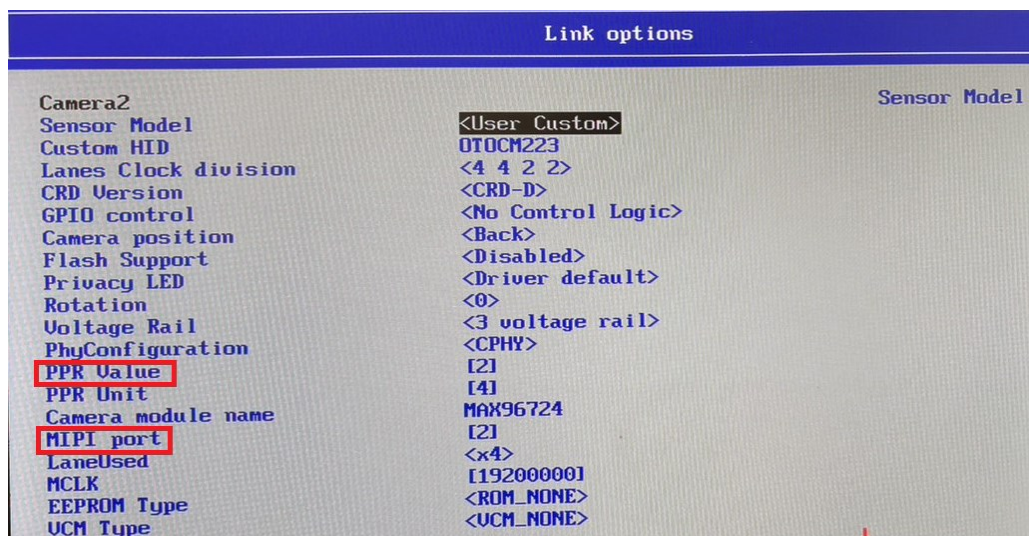
- while change new camera, do items from here:
 - (1) reboot the intel machine
 - enter BIOS
 - modify the HID for the specific sensor (refer to [Intel-MIPI-CSI-Camera-Reference-Driver/doc/isx031/userspace-gmsl.md at release/26Q1.2 · intel/Intel-MIPI-CSI-Camera-Reference-Driver · GitHub](#)) :
- UEFI Firmware Setting ->enter BIOS...-> Intel advanced menu->System Agent (SA) Configuration -> MIPI Camera Configuration -> Camera1&2 Link options
- ex: `otocam222 => OTOCM222`

otocam223 => OTOCM223
 otocam260 => OTOCM260 (support 3+3=6 otocam260isp)
 otocam271 => OTOCM271 (support 2+2=4 otocam271isp)
 otocam274 => OTOCM274 (support 4+4=8 otocam274isp)
 otocam276 => OTOCM276 (support 2+2=4 otocam276isp)
 otocam278 => OTOCM278 (support 4+4=8 otocam278isp)

BIOS Setting is based on “872110_IPU7 GMSL Enabling Guide_1.0.pdf” and fixed as following:

- (a) PPR Value = 2
- (b) MIPI port = 0 =>link 0
 port = 2 =>link 1





(2) enter ubuntu to preview image. (Do it every system start)

- \$cd /home/user/otocam-intel-driver-vx.xx.xx/script/
- \$sudo su
- \$source env.sh

- `./run_otocamxxx.sh` (for auto detect camera port, ctrl-c to stop steaming)

4.Steps to remove & insert driver (after remove & insert otocam):

- `$cd /home/user/otocam-intel-driver-vx.xx.xx/script/`
- `$rm_insert_drv.sh`